Configuration Checking with Rigorous Theoretical Foundation

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January 28, 2025

Abstract

1 Introduction

An important issue for local search is the cycling problem, i.e., revisiting a candidate solution that has been visited recently [MAK07]. Configuration checking (CC) is a strategy to reduce the cycling problem in local searches. The intuition behind this idea is that by reducing cycles on local structures of the candidate solution, we reduce cycles on the whole candidate solution [CS11]. However, the direct application of CC did not result in stochastic local search (SLS) algorithms that can compete with the current best SLS algorithms for SAT. In [CS12], a novel heuristic based on CC for SLS algorithms for SAT is proposed, demonstrating good performance. However, it remains unclear whether there exists a CC strategy that combines a solid theoretical foundation with consistently strong performance. In this paper, we first introduce a novel and intuitive CC strategy. We then provide an exact theoretical characterization of this strategy. Next, we demonstrate how to implement it efficiently. Finally, we validate its strong performance through experiments.

2 Checking the configurations of neighbors

Given a Conjunctive Normal Form (CNF) formula $F = C_1 \wedge C_2 \wedge \cdots \wedge C_m$ on a set of variables $\{X_1, \cdots, X_n\}$, the Boolean Satisfiability problem (SAT) consists in testing whether all clauses in F can be satisfied by some consistent assignment of truth values to variables. We use $V(F)$ to denote the set of all variables that appear in the formula F. A clause is a disjunction of literals, where a literal is either a variable X or its negation $\neg X$. We say a literal L occurs in a clause if this clause contains L. However, when we say a variable X occurs in a clause, we mean that this clause contains either X or $\neg X$. Two variables are neighbors iff they occur in at least one clause. Let $N(X) = \{Y | \ Y \in V(F), Y \neq X$ and Y occurs in at least one clause with $X\}$, which is the set of all neighboring variables of variable X. Let $N^+(X) = N(X) \cup \{X\}$. Let $\Delta(X)$ denote $|N(X)|$ and Δ denote the maximum $\Delta(X)$ over all $X \in V(F)$. We assume *w.l.o.g.* that $n \geq 2$ and $N(X_i) \neq \emptyset$ for each $i \in [n]$.

A (possibly partial) mapping $\alpha : V(F) \rightarrow \{0, 1\}$ is called an *assignment*. If α maps all variables to a Boolean value, it is called *complete*. Otherwise, it is called *partial*. For local search algorithms for SAT, a candidate solution is a complete assignment. For simplicity, we will refer to a complete assignment simply as an assignment. For any assignment S and any $i \in [n]$, let S^i denote the assignment obtained from S by flipping X_i.

Given a CNF formula F, consider the local search algorithm which begins by initializing a complete assignment for all variables and then iteratively flips variables one at a time to arrive at a solution for k-SAT. For any $t \geq 0$, let $s_0, s_1, s_2, \cdots, s_t = s$ denote the assignments of $V(F)$ in sequence where s_0 is the initial assignment and s_i is the assignment obtained after the i-th flip for each $i \in [t]$. The age of a variable is defined as the number of search steps that have occurred since the variable was last flipped.

In local search algorithms, the decisions are typically guided by the local configurations of the variables.

Definition 1. *Given any assignment s to $X_1, \cdots, X_n$ and any $i \in [n]$, let $s[X_i]$ denote the assignment of X_i in s. Similarly, for any subset $T \subseteq \{X_1, \cdots, X_n\}$, let $s[T]$ denote the vector consisting of truth values of all variables in T under s. The configuration $s(X_i)$ of X_i under s is defined as $s[N^+(X_i)]$.*

The main intuition of configuration check is to reduce cycles on the whole candidate solution by reducing cycles on local configurations of the candidate solution [CS11]. This intuition can be formulated as follows.

Observation 2. *Given any integers $t > r \geq 0$, any sequence of assignments $s_0, s_1, \cdots, s_t$ to $X_1, \cdots, X_n$, and any $i, j \in [n]$, if $s_t^i(X_j)$ is different from $s_r(X_j), s_{r+1}(X_j), \cdots, s_t(X_j)$, then s_t^i is different from $s_r, s_{r+1}, \cdots, s_t$.*

In previous algorithms [CS12, CLS15], when flipping a variable X_i, only the cycles on the local configuration of X_i is taken into consideration. This intuition can be formulated as follows.

Observation 3. *Given any integers $t > r \geq 0$, any sequence of assignments $s_0, s_1, \cdots, s_t$ to $X_1, \cdots, X_n$, and any $i \in [n]$, if $s_t^i(X_i)$ is different from $s_r(X_i), s_{r+1}(X_i), \cdots, s_t(X_i)$, then s_t^i is different from $s_r, s_{r+1}, \cdots, s_t$.*

Observation 3 represents only a partial realization of the configuration check concept. Compared to Observation 2, Observation 3 considers only the configuration of X_i. This restriction is overly strict, as demonstrated by the following lemma.

Lemma 4. *It is possible that $s_t^i(X_i) = s_r(X_i)$, but $s_t^i(X_j)$ is different from $s_r(X_j), \cdots, s_t(X_j)$ for some $i, j \in [n]$, integer $t > r \geq 0$ and assignments $s_r, s_{r+1}, \cdots, s_t$.*

Proof. Consider a CNF formula $F = (X_1 \vee X_2 \vee \neg X_3) \wedge (X_3 \vee X_4 \wedge X_5) \wedge (X_6 \vee X_7 \wedge X_8)$. Assume that $t = 2$ and the sequence of assignments are $s_0 = 00000000, s_1 = 01000000, s_2 = 01001000$. Thus, the assignment s_2^2 obtained from s_2 by flipping X_2 is $s_2^2 = 00001000$. The configuration $s_2^2(X_2)$, where X_1, X_2, X_3 are set as 000, is the same as $s_0(X_2)$. However, the configuration $s_2^2(X_3)$, where X_1, X_2, X_3, X_4, X_5 are set as 00001, is different from $s_0(X_3), s_1(X_3), s_2(X_3)$. □

2.1 q-flippable and e-flippable variables

To develop a fast local search algorithm, the configuration check must be implemented with limited resources. Therefore, it is impractical to verify whether a local configuration has ever been visited, as it is impossible to store all previously visited configurations within the constraints of limited storage. Instead of checking whether a local configuration has been visited at any point, we focus on determining whether it was visited most recently.

Definition 5 (Last configurations). *Given a CNF formula F, consider the local search algorithm which begins by initializing a complete assignment for all variables and then iteratively flips variables one at a time to arrive at a solution for k-SAT. Let $s_0, s_1, s_2, \cdots$ denote the assignments of $V(F) = \{X_1, \cdots, X_n\}$ in sequence where s_0 is the initial assignment and s_t is the assignment obtained after the t-th flip for each $t > 0$.*

For each $t \geq 0$, the assignment of F at step t is defined as s_t. For each $t > 0$, the last assignment of F at step t is defined as s_{t-1}. For each $t \geq 0$ and each $i \in [n]$, the last configuration of X_i at step t is $s_r(X_i)$, where r is the maximum number satisfying $0 \leq r < t$ and $s_r(X_i) \neq s_t(X_i)$. If such r does not exist, the last configuration of X_i at step t is undefined. The penultimate configuration of X_i at step t is $s_\ell(X)$, where ℓ is the maximum number satisfying $s_\ell(X_i) \neq s_r(X_i)$ and $s_r(X_i) \neq s_t(X_i)$ for some $0 \leq \ell < r < t$. If such ℓ does not exist, the penultimate configuration of X_i at step t is undefined.

For each $t > 0$ and $i \in [n]$, we say X_i is flipped at step t if $s_t = s^i_{t-1}$. To simplify, we assume that each variable is flipped at step 0.

We can define the eligible variables as follows.

Definition 6 (eligible variables). *Given any integer $t \geq 0$, any assignment s_t at step t, and any $i, j \in [n]$, we say X_i is eligible at step $t+1$ due to X_j if $s^i_t(X_j)$ is not the last configuration for X_j at step t. We note that $i = j$ is possible. For simplicity, we say (X_i, X_j) is eligible at step $t+1$ if X_i is eligible due to X_j.*

Intuitively, X_i is considered eligible if flipping it can result in a fresh local configuration for certain variables. We remark that (X_i, X_j) is eligible at step $t+1$ only if $X_j \in N^+(X_i)$, because $s^i_t(X_\ell) = s_t(X_\ell)$ for each $X_\ell \notin N^+(X_i)$. Note that whether a variable is eligible at the next step $t+1$ is determined by the assignment s_t at the current step t. Furthermore, we define the qualified variables as follows.

Definition 7 (qualified variables). *Given any integer $t \geq 0$, any assignment s_t at step t, and any $i \in [n]$, we say X_i is qualified at step $t+1$ if $s^i_t(X_i)$ is not the last configuration of X_i at step t.*

We remark that each variable is eligible and qualified at step 1, because the last configuration of each variable at step 0 is undefined.

The following lemma reveals the relationship between the eligible and qualified variables.

Lemma 8. *Given any integer $t \geq 0$ and any $i \in [n]$, X_i is qualified at step $t+1$ only if it is eligible.*

Proof. Let s_t be the assignment of $X_1, \cdots, X_n$ at step t. By X_i is qualified at step $t+1$, we have $s^i_t(X_i)$ is the last configuration of X_i at step t. Thus, (X_i, X_j) is eligible at step $t+1$. Hence, X_i is eligible at step $t+1$. □

For each variable X_i, the decision to flip X_i depends not only on whether flipping it changes the configuration of certain variables but also on the score associated with flipping X_i. To simplify our discussion, we assume the following condition on the score.

Condition 1 (Local score function). *For any assignment s and any variable X_i, the score for X_i in s is* $\mathrm{score}(F, X_i, s_t) \triangleq f(F, s^i) - f(F, s)$, *where the function f satisfies $f(F, s^i_1) - f(F, s_1) = f(F, s^i_2) - f(F, s_2)$ for any assignments s_1, s_2 where $s_1(X_i) = s_2(X_i)$.*

In a dynamic local search algorithm for SAT, a CNF formula F is combined with a weighting function w so that each clause $C \in F$ is associated with a positive integer number $w(C)$ as its weight. Condition 1 does not account for the influence of the weighting function because there are too many possible weighting functions, and their influence is extremely difficult to characterize.

Intuitively, Condition 1 ensures that the score of X_i in assignment s is determined solely by $s(X_i)$. A typical score of X_i in s is defined as the number of satisfied clauses under s^i minus the number of satisfied clauses under s. This score satisfies Condition 1. Any valid score, which effectively measures the impact of flipping variables, should have a similar effect to the difference in the number of satisfied clauses. Therefore, even for score functions that depend on the weighting function, our theorem based on Condition 1 still effectively captures the algorithm's primary behavior.

We define valuable variables as follows.

Definition 9. *Given any integer $t > 0$, any assignment s_t at step t, and any $i, j \in [n]$ where $X_i \in N^+(X_j)$, we say X_i is valuable at step $t+1$ if* $\mathsf{score}(F, X_i, s_t) > 0$. *Furthermore, we say (X_i, X_j) is valuable at step $t+1$ if and only if:*

1. *$i = j$ and* $\mathsf{score}(F, X_i, s_t) > 0$, *or*
2. *$i \neq j$ and the following conditions hold:*

$$\mathsf{score}(F, X_i, s_t) + \mathsf{score}(F, X_j, s_t^i) > 0, \qquad \mathsf{score}(F, X_j, s_t^i) > 0,$$
$$\mathsf{score}(F, X_j, s_t^i) > \mathsf{score}(F, X_j, s_t), \qquad \mathsf{score}(F, X_i, s_t^{i,j}) \leq 0.$$

Intuitively, (X_i, X_j) where $i \neq j$ is valuable only if the following conditions are satisfied.

- Flipping X_i and X_j results in a better assignment $s_t^{i,j}$ than s_t.
- Flipping X_j has a positive score after the flip of X_i. Otherwise, flipping X_j is unnecessary.
- Flipping X_i improves the score for flipping X_j. Otherwise, flipping X_i is not needed for flipping X_j.
- Flipping X_i and X_j consecutively does not lead to a valuable X_i. Otherwise, one can verify that X_i becomes qualified after the consecutive flipping of X_i and X_j. Combined with X_i is valuable, we have it should be flipped again, which is undesirable.

The flippable variables and pairs should simultaneously meet the requirements for the configuration and the score.

Definition 10. *Given any integer $t \geq 0$ and any $i, j \in [n]$ where $X_i \in N^+(X_j)$, X_i is called 1-step q-flippable at step $t+1$, or q-flippable for simplicity, if X_i is qualified and valuable. We say (X_i, X_j) is e-flippable at step $t+1$ if (X_i, X_j) is eligible and valuable.*

The following lemma is immediate by Definition 10.

Lemma 11. *Given any integer $t \geq 0$ and any $i \neq j \in [n]$, if (X_i, X_j) is e-flippable and X_i is flipped at step $t+1$, then X_j is q-flippable at step $t+2$.*

We use the standard interval notation. For example, $t \in (x_1, x_2)$ means $r_1 < t < r_2$ and $t \in [r_1, r_2]$ means $x_1 \leq t \leq x_2$. Given any variable X_i, we say X_i is flipped in (r_1, r_2) if and only if X_i is flipped at some step t where $r_1 < t < r_2$. The following definition is used in the definition of e-flippable variables.

Definition 12. *Given any integers $0 \leq r \leq t$ and any $i, j \in [n]$, assume that X_i is flipped at step r and never flipped in $(r, t]$. We say X_j is older than X_i at step $t+1$ if either X_j is flipped in $(r, t]$, or $r = 0$ and $j > i$.*

Intuitively, at step $t+1$, X_j is considered older than X_i if X_j was flipped after the last flip of X_i. If neither X_i nor X_j has been flipped since step 0, the variable with the larger index is regarded as older.

Now we can define the e-flippable variables by considering the effect of consecutively flipping two variables. If both pairs (X_i, X_j) and (X_j, X_i) are e-flippable, only one of the variables X_i or X_j needs to be flipped, as the other variable will become q-flippable after the flip due to Lemma 11. Thus, we only designate the older variable between X_i and X_j as e-flippable.

Definition 13. *Given any integer $t \geq 0$ and any $i, j \in [n]$ where $X_i \in N^+(X_j)$, we say X_i is e-flippable at step $t+1$ if there exists some j such that*

1. *(X_i, X_j) is e-flippable and (X_j, X_i) is not e-flippable, or*
2. *both (X_i, X_j) and (X_j, X_i) are e-flippable and X_j is not older than X_j.*

We note that X_i is e-flippable at step $t+1$ if (X_i, X_i) is e-flippable.

3 Effect of flipping e-flippable variables

Empirical studies have shown that restricting flips exclusively to 1-step q-flippable variables is overly limiting [CS12]. To address this, we relax the constraint by considering the effect of flipping e-flippable variables.

Given any two variables X_i and X_j where $X_j \in N(X_i)$, if consecutively flipping X_i and X_j results in a configuration of X_i or X_j that differs from its penultimate one, (X_i, X_j) should be considered qualified, as this reduces the cycles in the configuration of X_i or X_j. Here, we require $X_i \in N(X_j)$ because flipping X_i only affects the local configurations of its neighboring variables. Thus, we have the following definition on the qualified pair of variables.

Definition 14 (qualified pairs). *Given any integer $t \geq 0$, any assignment s_t at step t, and any $i, j \in [n]$ where $X_j \in N(X_i)$, we say (X_i, X_j) is qualified at step $t+1$ if either $s_t^{i,j}(X_i)$ is not the penultimate configuration of X_i at step t, or $S^{i,j}(X_j)$ is not the penultimate configuration of X_j at step t. Otherwise, we say (X_i, X_j) is unqualified at step $t+1$.*

Further more, we say (X_i, X_j) is q-flippable at step $t+1$ if (X_i, X_j) is qualified and valuable. We say X_i is 2-step q-flippable at step $t+1$ if there exists some $j \neq i$ such that

1. *(X_i, X_j) is q-flippable and (X_j, X_i) is not q-flippable, or*
2. *both (X_i, X_j) and (X_j, X_i) are q-flippable and X_j is not older than X_i.*

For simplify, we call such (X_i, X_j) unique q-flippable, or u-q-flippable. We say X_i is q-flippable if X_i is either 1-step or 2-step q-flippable.

By Definitions 9 and 14, we have the following lemma immediately.

Lemma 15. *Assume Condition 1 is satisfied. Given any integer $t \geq 0$, any $i, j \in [n]$ where $X_i \in N(X_j)$ and (X_i, X_j) is unqualified and valuable at step $t+1$, assume X_i is older than X_j and the last flip of X_i is at some step $r < t$, then we have*

- $\mathrm{score}(F, X_i, s_{r-1}) \leq 0$,
- $s_{r-1}(X_i) = s_t^{i,j}(X_i), s_{r-1}(X_j) = s_t^{i,j}(X_j)$, *each $X_k \in N(X_i) \cup N(X_j) \setminus \{X_i, X_j\}$ is not flipped in $[r, t]$.*
- $\mathrm{score}(F, X_i, s_{r-1}) + \mathrm{score}(F, X_j, s_{r-1}^i) < 0$, $\mathrm{score}(F, X_i, s_t) + \mathrm{score}(F, X_j, s_t^i) > 0$.

Proof. By (X_i, X_j) is unqualified at step $t+1$, we have $s_t^{i,j}(X_i), s_t^{i,j}(X_j)$ are the penultimate configurations of X_i and X_j at step t, respectively. By that the penultimate configurations of X_i and X_j at step t exist and that X_i is older than X_j and the last flip of X_i is at step r, we have $r \geq 1$ and $s_{r-1}(X_i), s_{r-1}(X_j)$ are the penultimate configurations of X_i and X_j at step t, respectively. Hence, $s_{r-1}(X_i) = s_t^{i,j}(X_i), s_{r-1}(X_j) = s_t^{i,j}(X_j)$.

Assume for contradiction that some variable in $N(X_i) \cup N(X_j) \setminus \{X_i, X_j\}$ is flipped in $[r, t]$. By X_i is older than X_j and the last flip of X_i is at step r, we have (X_i, X_j) is qualified at step $t+1$, no matter $r > 0$ or $r = 0$. This is contradictory with (X_i, X_j) is unqualified at step $t+1$.

By (X_i, X_j) is valuable at step $t+1$, we have $\mathrm{score}(F, X_i, s_t^{i,j}) \leq 0$. Combined with $s_{r-1}(X_i) = s_t^{i,j}(X_i)$ and Condition 1, we have $\mathrm{score}(F, X_i, s_{r-1}) \leq 0$. Similarly, by (X_i, X_j) is valuable at step $t+1$, we have

$$\mathrm{score}(F, X_i, s_t) + \mathrm{score}(F, X_j, s_t^i) > 0.$$

Meanwhile, by $s_{r-1}(X_i) = s_t^{i,j}(X_i)$ we have $s_{r-1}^i(X_i) = s_t^j(X_i)$. Combined with Condition 1, we have

$$\text{score}(F, X_i, s_{r-1}) = f(F, s_{r-1}^i) - f(F, s_{r-1}) = f(F, s_t^j) - f(F, s_t^{i,j}).$$

Similarly, by $s_{r-1}(X_j) = s_t^{i,j}(X_j)$ we have $s_{r-1}^{i,j}(X_j) = s_t(X_j)$ and $s_{r-1}^i(X_j) = s_t^j(X_j)$. Combined with Condition 1, we have

$$\text{score}(F, X_j, s_{r-1}^i) = f(F, s_{r-1}^{i,j}) - f(F, s_{r-1}^i) = f(F, s_t) - f(F, s_t^j).$$

Combining above three inequalities, we have

$$\begin{aligned}\text{score}(F, X_i, s_{r-1}) + \text{score}(F, X_j, s_{r-1}^i) &= f(F, s_t^j) - f(F, s_t^{i,j}) + f(F, s_t) - f(F, s_t^j) \\ &= -(f(F, s_t^{i,j}) - f(F, s_t^i) + f(F, s_t^i) - f(F, s_t)) \\ &= -(\text{score}(F, X_i, s_t) + \text{score}(F, X_j, s_t^i)) < 0.\end{aligned}$$

This finishes the proof of the lemma. □

By Lemma 15, we make the following observations. First, the flip of X_i at step r is a random attempt, not a greedy choice. Second, for the variables in $N(X_i) \cup N(X_j)$, only X_i and X_j are flipped in $[r, t]$. If we flip X_i and X_j at steps $t+1$ and $t+2$, the assignments of X_i and X_j are reversed, and the configurations of X_i and X_j are recovered. Finally, flipping X_i at step r does not result in a better assignment than s_r, even if a neighboring variable of X_i is also flipped. In contrast, reversing X_i and X_j in steps $t+1$ and $t+2$ results in the assignment $s_t^{i,j}$, which is better than s_t. In summary, although flipping X_i and X_j consecutively does not produce a new local configuration for either X_i or X_j, it corrects the erroneous flip of X_i at step r. Thus, we have the following definition on the reversible variables.

Definition 16 (reversible variables). *Given any integer $t > 0$, any assignment s_t at step t, and any $i, j \in [n]$ where $X_j \in N(X_i)$, we say X_i is reversible due to X_j if (X_i, X_j) is unqualified and valuable and X_i is older than X_j. If X_i is reversible due to any $X_j \in N(X_i)$, we say X_i is reversible.*

To simplify the discussion, we further assume the following condition, which is satisfied by all the algorithms in the paper.

Condition 2 (Greedy choice). *For any integer $t \geq 0$, any assignment s_t at step t, and any variable X_i, if there exists some qualified X_j at step $t+1$ where $\text{score}(F, X_j, s_t) > 0$ and $\text{score}(F, X_j, s_t) > \text{score}(F, X_i, s_t)$, then X_i is not flipped at step $t+1$.*

Now we can state the main result of this section.

Theorem 17. *Assume Conditions 1 and 2 are satisfied. For any integer $t \geq 0$ and any $i \in [n]$, X_i is e-flippable at step $t+1$ if and only if X_i is either q-flippable or reversible at step $t+1$.*

Theorem 17 provides an exact characterization of the set of e-flippable variables, which is precisely the union of the q-flippable variables and the reversible variables. Thus, when an e-flippable variable is flipped, either a better assignment can be achieved in two steps, while reducing the cycle in local configurations, or an erroneous flip can be corrected in two steps.

A surprising conclusion of Theorem 17 is that each q-flippable variable is also e-flippable. Note that even if (X_i, X_j) is q-flippable, it is still possible for (X_i, X_j) to not be e-flippable, since X_i may be the last

flipped variable in $N^+(X_j)$ at step $t+1$. The key observation is that if X_i is 1-step q-flippable at step $t+1$, or if (X_j, X_i) is q-flippable and X_i is older than X_j, then X_i is immediately e-flippable at step $t+1$.

Thus, it suffices to consider the case where (X_i, X_j) is q-flippable, but neither X_i is a 1-step q-flippable nor (X_j, X_i) is q-flippable. Suppose X_i is flipped at some step $r \leq t$ and never flipped in $(r, t]$, and that X_i is older than X_j at step $t+1$. In Lemma 19, we show that no variable in $N(X_i)$ is flipped in $(r, t]$. Otherwise, (X_j, X_i) would be valuable at step $t+1$ according to Condition 1 and (X_i, X_j) is valuable, which contradicts the assumption that (X_i, X_j) is q-flippable and (X_j, X_i) is not q-flippable.

Since no variable in $N(X_i)$ is flipped in $(r, t]$, we conclude that some variable in $N(X_j)$ must have been flipped in $(r, t]$. Otherwise, (X_i, X_j) would be unqualified at step $t+1$, which contradicts the assumption that (X_i, X_j) is q-flippable. Therefore, since some variable in $N(X_j)$ must have been flipped in $(r, t]$, it follows that (X_i, X_j) is eligible at step $t+1$. It is then straightforward to verify that (X_i, X_j) is e-flippable.

The following two lemmas are used in the proof of Theorem 17.

Lemma 18. *Given any integer $t \geq 0$ and any X_i reversible at step $t+1$, we have X_i is also e-flippable at step $t+1$,*

Proof. Let s_t be the assignment of F at step t. By X_i is reversible at step $t+1$, we have there exists some $X_j \in N(X_i)$ where (X_i, X_j) is unqualified and valuable and X_i is older than X_j. By (X_i, X_j) is unqualified, we have $s_t^{i,j}(X_j)$ is the penultimate configuration of X_j at step $t+1$. Combined with X_i is older than X_j, we have $s_t^i(X_j)$ is not the last configuration of X_j at step $t+1$. Thus, (X_i, X_j) is eligible at step $t+1$. Combined with that (X_i, X_j) is valuable at step $t+1$, we have (X_i, X_j) is e-flippable. Combined with that X_i is older than X_j, we have X_i is also e-flippable at step $t+1$. □

Lemma 19. *Assume Conditions 1 and 2 are satisfied. Given any CNF formula F, any integer $t > 0$ and any (X_i, X_j) q-flippable at step t, assume that X_i is not 1-step q-flippable. If (X_j, X_i) is q-flippable and X_j is older than X_i at step $t+1$, then (X_j, X_i) is e-flippable at step $t+1$. If (X_j, X_i) is not q-flippable at step $t+1$, then (X_i, X_j) is e-flippable at step $t+1$.*

Proof. Let s_t be the assignment of F at step t. Suppose that X_i is flipped at some step r and never flipped in $(r, t]$, X_j is flipped at some step ℓ and never flipped in $(\ell, t]$. By (X_i, X_j) is q-flippable at step t, we have $X_j \in N(X_i)$. First, consider the case where (X_j, X_i) is q-flippable and X_j is older than X_i at step $t+1$. There are two possibilities:

- $r = \ell = 0$. We have (X_j, X_i) is eligible immediately. In addition, by (X_j, X_i) is q-flippable, we have (X_j, X_i) is valuable. Combined with (X_j, X_i) is eligible, we have (X_j, X_i) is e-flippable at step $t+1$.
- $r > \ell \geq 0$. By $X_j \in N(X_i)$ and $r > \ell$, we have $s_t^j(X_i)$ is not the last configuration of X_i at step t. Therefore, (X_j, X_i) is eligible at step $t+1$. Moreover, by (X_j, X_i) is q-flippable at step $t+1$, we have (X_j, X_i) is valuable. Combined with (X_j, X_i) is eligible at step t, we have (X_j, X_i) is e-flippable at step $t+1$.

In summary, we always have (X_j, X_i) is e-flippable at step $t+1$.

Next, we consider the case where (X_j, X_i) is not q-flippable at step $t+1$. By (X_i, X_j) is q-flippable at step $t+1$, we have (X_i, X_j) is valuable at step $t+1$. Thus, to show (X_i, X_j) is e-flippable at step $t+1$, it is sufficient to show (X_i, X_j) is eligible at step $t+1$. In the following, we prove (X_i, X_j) is eligible at step $t+1$ by considering three separate cases:

- $r < \ell$. Thus, $s_t^i(X_j)$ is not the last configuration for X_j at step t. Hence, (X_i, X_j) is eligible at step $t+1$.

- $r = \ell = 0$. We have (X_i, X_j) is eligible at step $t+1$ immediately.
- $r > \ell$ and (X_j, X_i) is not q-flippable at step $t+1$. Let s_{r-1} be the assignment of F at step $r-1$. We claim that some variable in $N(X_j)$ is flipped in $(r, t]$. Then $s_t^i(X_j)$ is not the last configuration for X_j at step t. Therefore, we have (X_i, X_j) is eligible at step $t+1$.

  In the next, we prove the claim that some variable in $N(X_j)$ is flipped in $(r, t]$. We first claim that no variable in $N(X_i)$ is flipped in $(r, t]$ and prove that some variable in $N(X_j)$ is flipped in $(r, t]$. Assume for contradiction that no variable in $N(X_j)$ is flipped in $(r, t]$. If $\ell > 0$, by no variable in $N(X_i)$ or $N(X_j)$ is flipped in $(r, t]$, we have $s_t^{i,j}(X_i), s_t^{i,j}(X_i)$ are the penultimate configurations of X_i and X_j at step t, respectively. Thus, (X_i, X_j) is not qualified at step $t+1$, which is contradictory with (X_i, X_j) is q-flippable at step $t+1$. Thus, we have $\ell = 0$. Therefore, X_j is qualified at step r. Moreover, by no variable in $N(X_j)$ is flipped in $(r, t]$, we have $s_t^i(X_j)$ is last the configuration for X_j at step t. Formally, $s_t^i(X_j) = s_{r-1}(X_j)$. In addition, by (X_i, X_j) q-flippable at step t, we have (X_i, X_j) valuable at step t. Thus,

$$\text{score}(F, X_i, s_t) + \text{score}(F, X_j, s_t^i) > 0.$$

  We have

$$\text{score}(F, X_j, s_t^i) > -\text{score}(F, X_i, s_t) = -(f(F, s_t^i) - f(F, s_t)) = \text{score}(F, X_i, s_t^i).$$

  In addition, by $s_t^i(X_j) = s_{r-1}(X_j)$ and Condition 1, we have $\text{score}(F, X_j, s_{r-1}) = \text{score}(F, X_j, s_t^i)$. Thus,

$$\text{score}(F, X_j, s_{r-1}) > \text{score}(F, X_i, s_t^i).$$

  Moreover, by no variable in $N(X_i)$ is flipped in $(r, t]$, we have $s_t(X_i) = s_{r-1}^i(X_i)$. Thus, we have $s_t^i(X_i) = s_{r-1}(X_i)$ and $\text{score}(F, X_i, s_t^i) = \text{score}(F, X_i, s_{r-1})$. Combined with $\text{score}(F, X_j, s_{r-1}) > \text{score}(F, X_i, s_t^i)$, we have

$$\text{score}(F, X_j, s_{r-1}) > \text{score}(F, X_i, s_{r-1}). \tag{1}$$

  In addition, also by (X_i, X_j) valuable at step t, we have $\text{score}(F, X_j, s^i) > 0$. Combined with $s^i(X_j) = s_{r-1}(X_j)$ and Condition 1, we have $\text{score}(F, X_j, s_{r-1}) > 0$. Combined with (1), X_j is qualified at step r, and Condition 2, we have X_i is not flipped at step r, a contradiction. Thus, some variable in $N(X_j)$ is flipped in $(r, t]$.

  At last, we prove the claim that no variable in $N(X_i)$ is flipped in $(r, t]$. Assume for contradiction that some variable in $N(X_i)$ is flipped in $(r, t]$. Thus, we have X_i is qualified at step $t+1$. Combined with X_i is not 1-step q-flippable, we have $\text{score}(F, X_i, s_t) \le 0$. In addition, by (X_i, X_j) is q-flippable at step $t+1$, we have (X_i, X_j) is valuable at step $t+1$. Thus, $\text{score}(F, X_i, s_t) + \text{score}(F, X_j, s_t^i) > 0$. We have

$$\begin{aligned}
\text{score}(F, X_j, s_t) + \text{score}(F, X_i, s_t^j) &= f(F, s_t^j) - f(F, s_t) + f(F, s_t^{i,j}) - f(F, s_t^j) \\
&= f(F, s_t^i) - f(F, s_t) + f(F, s_t^{i,j}) - f(F, s_t^i) \\
&= \text{score}(F, X_i, s_t) + \text{score}(F, X_j, s_t^i) > 0.
\end{aligned}$$

  Combined with $\text{score}(F, X_j, s_t) \le 0$, we have $\text{score}(F, X_i, s_t^j) > 0$. Moreover, by $\text{score}(F, X_i, s_t) \le 0$ and $\text{score}(F, X_i, s_t) + \text{score}(F, X_j, s_t^i) > 0$, we have $\text{score}(F, X_j, s_t^i) > 0$. Thus,

$$\text{score}(F, X_j, s_t^{i,j}) = f(F, s_t^i) - f(F, s_t^{i,j}) = -\text{score}(F, X_j, s_t^i) < 0.$$

In summary, we have

$$\text{score}(F, X_i, s_t^j) > 0, \quad \text{score}(F, X_i, s_t^j) > \text{score}(F, X_i, s_t)$$
$$\text{score}(F, X_j, s_t^{i,j}) \leq 0, \quad \text{score}(F, X_j, s_t) + \text{score}(F, X_i, s_t^j) > 0.$$

Thus, (X_j, X_i) is valuable at step t. Moreover, by (X_i, X_j) is q-flippable at step $t+1$, we have (X_i, X_j) is qualified at step $t+1$. Equivalently, (X_j, X_i) is qualified at step $t+1$. Combined with (X_j, X_i) is valuable at step $t+1$, we have (X_j, X_i) is q-flippable at step $t+1$, a contradiction. Therefore, the claim that no variable in $N(X_i)$ is flipped in $(r, t]$ is proved. Hence, we have some variable in $N(X_j)$ is flipped in $(r, t]$ and (X_i, X_j) is eligible at step $t+1$.

In summary, we always have (X_i, X_j) is eligible at step $t+1$. Thus, (X_i, X_j) is e-flippable at step $t+1$. The lemma is proved. □

Now we can prove Theorem 17.

Proof of Theorem 17. Let s_t be the assignment of F at step t. At first, we assume X_i is e-flippable at step $t+1$ and show X_i is either q-flippable or reversible. By X_i is e-flippable at step $t+1$, we have there are three possibilities:

- (X_i, X_i) is e-flippable at step $t+1$. We have (X_i, X_i) is eligible and valuable at step $t+1$. By (X_i, X_i) is eligible at step $t+1$ and Definition 6, we have $S^i(X_i)$ is not the last configuration for X_i at step t. Thus, X_i is qualified at step $t+1$. In addition, by (X_i, X_i) is valuable at step $t+1$, we have $\text{score}(F, X_i, s_t) > 0$. Combined with that X_i is qualified at step $t+1$, we have X_i is q-flippable.
- There exists some X_j such that either (X_i, X_j) is e-flippable and (X_j, X_i) is not e-flippable, or both (X_i, X_j) and (X_j, X_i) are e-flippable and X_i is older than X_j. We consider two separate cases.
  - (X_i, X_j) is unqualified at step $t+1$. Assume that X_i is flipped at some $r \leq t$ and never flipped in $(r, t]$, X_j is flipped at some $\ell \leq t$ and never flipped in $(\ell, t]$. Combined with (X_i, X_j) is unqualified at step $t+1$, we have $r > 0$ and $\ell > 0$. Otherwise, $s_t^{i,j}(X_j)$ is not the penultimate configuration of X_i at step t. Assume $r > \ell$ for contradiction, by (X_i, X_j) is unqualified, we have $s_t^{i,j}(X_j)$ is the penultimate configuration of X_j at step $t+1$. Combined with $r > \ell$, we have $s_t^i(X_j)$ is the last configuration of X_j at step $t+1$, which is contradictory with (X_i, X_j) is eligible at step $t+1$. Thus, we have $r < \ell$. Combined with that (X_i, X_j) is unqualified and valuable, we have X_i is reversible immediately.
  - (X_i, X_j) is qualified at step $t+1$. By (X_i, X_j) is e-flippable at step $t+1$, we have (X_i, X_j) is valuable at step $t+1$. Combined with that (X_i, X_j) is qualified at step $t+1$, we have (X_i, X_j) is q-flippable immediately. If (X_j, X_i) is e-flippable and X_i is older than X_j, by (X_i, X_j) is q-flippable, we have X_i is q-flippable. Otherwise, (X_j, X_i) is not e-flippable. We have either (X_j, X_i) is not eligible, or (X_j, X_i) is not valuable. If (X_j, X_i) is not valuable, we have (X_i, X_j) is not q-flippable immediately. Combined with (X_i, X_j) is q-flippable, we have X_i is q-flippable. Otherwise, we have (X_j, X_i) is not eligible. Thus, $s_t^j(X_i)$ is the last configuration of X_i at step $t+1$. We have X_j is older than X_i at step t. Combined with (X_i, X_j) is q-flippable, we also have X_i is q-flippable. In summary, we always have X_i is q-flippable.

In the following, we assume X_i is either q-flippable or reversible at step $t+1$ and show X_i is e-flippable. If X_i is reversible at step $t+1$, we have X_i is e-flippable immediately by Lemma 18. If X_i is q-flippable at

step $t+1$, we have X_i is either 1-step q-flippable or 2-step q-flippable at step $t+1$. If X_i is 1-step q-flippable, we have X_i is qualified at step $t+1$ and $\text{score}(F, X_i, s_t) > 0$. By X_i is qualified at step $t+1$, we have $s_t^i(X_i)$ is not the last configuration of X_i at step t. Thus, (X_i, X_i) is eligible at step $t+1$. Combined with $\text{score}(F, X_i, s_t) > 0$, we have (X_i, X_i) is e-flippable at step $t+1$. Thus, X_i is e-flippable at step $t+1$. If X_i is 2-step q-flippable, we have there exists some $j \neq i$ such that either (X_i, X_j) is q-flippable and (X_j, X_i) is not q-flippable, or both (X_i, X_j) and (X_j, X_i) are q-flippable and X_i is older than X_j.

- If (X_i, X_j) is q-flippable and (X_j, X_i) is not q-flippable, we have (X_i, X_j) is qualified, valuable and (X_j, X_i) is not valuable. Thus, (X_j, X_i) is not e-flippable. In addition, by Lemma 19, we have (X_i, X_j) is e-flippable. Combined with (X_j, X_i) is not e-flippable, we have X_i is e-flippable at step $t+1$.
- If both (X_i, X_j) and (X_j, X_i) are q-flippable and X_i is older than X_j, by Lemma 19, we have (X_i, X_j) is e-flippable at step $t+1$. If (X_j, X_i) is not e-flippable at step $t+1$, we have X_i is e-flippable at step $t+1$. Otherwise, (X_j, X_i) is also e-flippable at step $t+1$. Combined with X_i is older than X_j, we have X_i is e-flippable at step $t+1$. In summary, we always have X_i is e-flippable at step $t+1$.

In summary, X_i is e-flippable at step $t+1$.

□

4 Fast implement of flipping e-flippable variables

In this section, we demonstrate how to efficiently implement the strategy for flipping the e-flippable variables. According to Theorem 17, flipping the e-flippable variables requires flipping the 1-step q-flippable variables, the 2-step q-flippable variables, and the reversible variables. It is worth noting that identifying the 2-step q-flippable and reversible variables is significantly more time-consuming than identifying the 1-step q-flippable variables. This is because the 2-step q-flippable and reversible variables account for the effects of consecutively flipping two neighboring variables, whereas the 1-step q-flippable variables consider only the effect of flipping a single variable. In the following, we will demonstrate an efficient method for identifying the 2-step q-flippable and reversible variables.

4.1 The overall algorithm

Our algorithm for flipping e-flippable variables follows the same framework as the Swcca algorithm, which flips 1-step q-flippable variables.

The Swcca algorithm is an efficient algorithm of k-SAT. To focus on the essential part of the Swcca algorithm, we only present the pseudo-code of its pickVar function, shown in Algorithm 1, as the initialization is trivial: generating a random complete assignment s, initiating all clause weights as 1 and computing scores of variables accordingly. In all algorithms, ties are always broken in favor of the oldest variable.

The following definitions are used in the pickVar function. A significant decreasing (SD) variable is a variable with $\text{score}(x) > g$, where g is a positive integer large enough.

Instead of the 1-step q-flippable and SD variables used in the pickVar function, we aim to flip the e-flippable variables as in the pickEligibleVar function, shown in Algorithm 2. In contrast to the SD variables, which only account for the effect of a single flip, the 2-step q-flippable and reversible variables consider the effects of consecutively flipping two neighboring variables.

The method for maintaining the 1-step q-flippable variables follows the approach in [CS12]. The scoring method used here is also the same as that in [CS12].

Algorithm 1: pickVar-function

```
//greedy model
if any variable is 1-step q-flippable then return such a variable with the largest score;
if any variable is SD then return such a variable with the largest score;
//diversification model
update clause weights;
pick a random unsatisfied clause c;
return the oldest variable in c;
```

Algorithm 2: pickEligibleVar-function

```
//greedy model
if any variable is 1-step q-flippable then return such a variable with the largest score;
if any variable is 2-step q-flippable then return such a variable with the largest score;
if any variable is reversible then return such a variable with the largest score;
//diversification model
update clause weights;
pick a random unsatisfied clause c;
return the oldest variable in c;
```

Below, we provide the details of the method for maintaining the 2-step q-flippable and reversible variables. To efficiently find the variable with the largest score, we use the difference in the number of satisfied clauses as the score. This serves as a good approximation of the original score and can be computed efficiently.

4.2 The critical and non-critical pairs

Given any 3-SAT, if a clause shares at least two variables with other clauses, we call it critical. Otherwise, we call it non-critical. For any pair (X_i, X_j), we call it critical if there exists some critical clause containing X_i and X_i simultaneously. Otherwise, we call it non-critical. For any variable X_i, if there exists some X_j such that (X_i, X_j) is critical, we call X_i critical. Otherwise, we call it non-critical. Our method for maintaining the 2-step q-flippable and reversible variables is based on the following two observations.

Observation 20. *For any random 3-SAT with m clauses and n variables, the expected number of critical clauses is no more than $18m^2/n^2$.*

Proof. For any fixed clause, the probability that it shares at least two variables with other clauses is no more than $3 \times 6 \times m/n^2$. Thus, the expected number of critical clause is no more than $18m^2/n^2$. □

By Observation 20, for any random 3-SAT where m/n is a constant, the expected number of critical variables is also a constant.

For any assignment s and any variable X_i, let $N(F, X_i, s)$ denote the difference between the number of satisfied clauses under s^i and the number of satisfied clauses under s. In addition, for any variable X_j where $j \neq i$, let $N(F, X_i, X_j, s)$ denote the difference between the number of satisfied clauses under $s^{i,j}$ and the number of satisfied clauses under s. For typical 3-SAT instance, $N(F, X_i, X_j, s)$ is generally no more

than a few dozen. Given any clause C and any assignment s, we let $C(s) = 1$ if C is satisfied under s and let $C(s) = 0$, otherwise. Then we have the following observation.

Observation 21. *Suppose the difference in the number of satisfied clauses is used as the score function. Given any 3-SAT formula F, any integer $t \geq 0$, any assignment s_t at step t, and any non-critical (X_i, X_j) of F, (X_i, X_j) is valuable at step $t+1$ if and only if*

- *there exists exact one clause C such that the variables X_i and X_j occur in C together.*
- *exact one variable satisfies C under s_t, and this variable is either X_i or X_j.*
- *either $N(F, X_j, s_t) \geq 0, N(F, X_i, s_t) \geq 0$ or $N(F, X_j, s_t) \geq 1, N(F, X_i, s_t) = -1$.*

Proof. Assume that (X_i, X_j) is valuable at step $t+1$. By (X_i, X_j) is non-critical, there is at most one clause in which the variables X_i and X_j occur together. By (X_i, X_j) is valuable, we have $N(F, X_j, s_t^i) > N(F, X_j, s_t)$. Thus, there is at least one clause in which X_i and X_j occur together. Therefore, there exists exact one clause C in which X_i and X_j occur together. Meanwhile, by $N(F, X_j, s_t^i) > N(F, X_j, s_t)$ and C is the unique clause in which X_i and X_j occur together, we have either $C(s_t^j) - C(s_t) = -1$ and $C(s_t^{i,j}) - C(s_t^i) = 0$, or $C(s_t^j) - C(s_t) = 0$ and $C(s_t^{i,j}) - C(s_t^i) = 1$. If $C(s_t^j) - C(s_t) = -1$, we have X_j is the unique variable satisfying C under s_t. Otherwise, we have $C(s_t^j) - C(s_t) = 0$ and $C(s_t^{i,j}) - C(s_t^i) = 1$. Thus, $C(s_t^i) = 0$, $C(s_t^{i,j}) = 1$. Therefore, $C(s_t^j) = 1$. Combined with $C(s_t^j) - C(s_t) = 0$, we have $C(s_t) = 1$. Combined with $C(s_t^i) = 0$, we have X_i is the unique variable satisfying C under s_t. In summary, we have exact one variable satisfies C under s_t, and this variable is either X_i or X_j. Moreover, by (X_i, X_j) is valuable, we have $N(F, X_i, X_j, s_t) > 0, N(F, X_j, s_t^i) > 0$ and $N(F, X_i, s_t^{i,j}) \leq 0$. By $N(F, X_j, s_t^i) > 0$ and C is the unique clause in which X_i and X_j occur together, we have $N(F, X_j, s_t) \geq N(F, X_j, s_t^i) - 1 \geq 0$. By $N(F, X_i, s_t^{i,j}) \leq 0$, we have $N(F, X_i, s_t^j) \geq 0$. Combined with C is the unique clause in which X_i and X_j occur together, we have $N(F, X_i, s_t) \geq N(F, X_i, s_t^j) - 1 \geq -1$. Moreover, by C is the unique clause in which X_i and X_j occur together, we have

$$N(F, X_j, s_t) + N(F, X_i, s_t) \geq N(F, X_j, s_t^i) + N(F, X_i, s_t) - 1 = N(F, X_i, X_j, s_t) - 1 \geq 0.$$

Combined with $N(F, X_j, s_t) \geq 0$ and $N(F, X_i, s_t) \geq -1$, we have either $N(F, X_j, s_t) \geq 0, N(F, X_i, s_t) \geq 0$ or $N(F, X_j, s_t) \geq 1, N(F, X_i, s_t) = -1$.

Assume that C is the unique clause in which X_i and X_j occur together, either X_i or X_j is the only variable satisfying C, and either $N(F, X_j, s_t) \geq 0, N(F, X_i, s_t) \geq 0$ or $N(F, X_j, s_t) \geq 1, N(F, X_i, s_t) = -1$. We claim that $N(F, X_j, s_t^i) = N(F, X_j, s_t) + 1$ and $N(F, X_i, s_t^j) = N(F, X_i, s_t) + 1$. By $N(F, X_j, s_t) \geq 0$ and $N(F, X_j, s_t^i) = N(F, X_j, s_t) + 1$, we have $N(F, X_j, s_t^i) \geq 1$. By $N(F, X_i, s_t) \geq -1$ and $N(F, X_i, s_t^j) = N(F, X_i, s_t) + 1$, we have $N(F, X_i, s_t^j) \geq 0$. If $N(F, X_j, s_t) \geq 0$ and $N(F, X_i, s_t) \geq 0$, by $N(F, X_j, s_t^i) = N(F, X_j, s_t) + 1$ we have

$$N(F, X_i, X_j, s_t) = N(F, X_j, s_t^i) + N(F, X_i, s_t) = N(F, X_j, s_t) + N(F, X_i, s_t) + 1 \geq 1.$$

If $N(F, X_j, s_t) \geq 1$ and $N(F, X_i, s_t) = -1$, by $N(F, X_j, s_t^i) = N(F, X_j, s_t) + 1$ we also have $N(F, X_i, X_j, s_t) \geq 1$. In summary, we always have $N(F, X_i, X_j, s_t) \geq 1$. Combined with $N(F, X_i, s_t^j) \geq 0$, $N(F, X_j, s_t^i) \geq 1$, $N(F, X_j, s_t^i) = N(F, X_j, s_t) + 1$, we have (X_i, X_j) is valuable. In the following, we prove the claim $N(F, X_j, s_t^i) = N(F, X_j, s_t) + 1$ and $N(F, X_i, s_t^j) = N(F, X_i, s_t) + 1$ by considering two different cases separately.

- Assume X_i is the only variable satisfying C. We have $C(s_t^j) - C(s_t) = 0$ and $C(s_t^{i,j}) - C(s_t^i) = 1$. Therefore, $N(F, X_j, s_t^i) = N(F, X_j, s_t) + 1$. Similarly, we also have $C(s_t^i) - C(s_t) = -1$ and $C(s_t^{i,j}) - C(s_t^j) = 0$. Thus, $N(F, X_i, s_t^j) = N(F, X_i, s_t) + 1$.

- Assume X_j is the only variable satisfying C. We have $C(s_t^j) - C(s_t) = -1$ and $C(s_t^{i,j}) - C(s_t^i) = 0$. Therefore, $N(F, X_j, s_t^i) = N(F, X_j, s_t)+1$. Similarly, we also have $C(s_t^i)-C(s_t) = 0$ and $C(s_t^{i,j})-C(s_t^j) = 1$. Thus, $N(F, X_i, s_t^j) = N(F, X_i, s_t) + 1$.

Thus, the claim is proved and (X_i, X_j) is valuable. This finishes the proof of the lemma. □

By Observation 21, when checking for q-flippable and reversible pairs that are non-critical, it suffices to consider only those pairs (X_i, X_j) where X_i, X_j occur together in some non-critical clause.

4.3 Dynamic data structures

Recall that $F = C_1 \wedge C_2 \wedge \cdots \wedge C_m$. For each variable X_i, we record the clauses C in which the variable X_i occurs. We also maintain $N(F, X_i, s)$ where s is the current assignment. One can verify that $N(F, X_1, s), \cdots, N(F, X_n, s)$ can be maintained within $O(\Delta)$ time for each flip.

4.3.1 Data structure for checking qualification and reversibility

For each variable X_i and any step $t > 0$, we use $L_1(X_i)$ to record the last flipped variable in $N^+(X_i)$ in step $(0, t]$ and $L_2(X_i)$ to record the penultimate flipped variable in $N^+(X_i)$ in step $(0, t]$. Initially, $L_1(X_i)$ and $L_2(X_i)$ are set as null. It is easy to verify that $L_1(X_1), \cdots, L_1(X_n)$, $L_2(X_1), \cdots, L_2(X_n)$ can be maintained within $O(\Delta)$ time for each flip. Moreover, for each pair (X_i, X_j), it is easy to check whether (X_i, X_j) is qualified and whether X_i is older than X_j by checking the values of $L_1(X_i), L_2(X_1), L_1(X_j), L_2(X_j)$. Thus, one can easily determine whether (X_i, X_j) is qualified and reversible by checking $L_1(X_i), L_2(X_i), L_1(X_j), L_2(X_j)$.

4.3.2 Data structure for checking critical pairs

Given a CNF formula F, we use a list CPair to record the list of critical pairs. The establishment of CPair is as follows. Initially, CPair is set as null. We traverse all the clauses of F. For each clause C, we add all the variable pairs (X_i, X_j) contained in C to CPair. Here, (X_i, X_j) and (X_j, X_i) are regarded as the same pair and we only add the pair (X_i, X_j) where $i < j$ into CPair. In the next, by sorting all the pairs in CPair and check whether a pair is the same with its successor, one can obtain the list of critical pairs easily. For each critical variable X_i, we also use a list $LCP(X_i)$ to record the critical pairs containing X_i. For each critical pair (X_i, X_j), we use a list $LCC(X_i, X_j)$ to record all the critical clauses in which X_i and X_j occur together.

We maintain the list LCQ of critical and u-q-flippable pairs (X_i, X_j), and their scores $N(F, X_i, X_j, s)$ where s is the current assignment. For each critical variable X_i, once any variable in $N^+(X_i)$ is flipped, for each pair (X_i, X_j) in $LCP(X_i)$, one can check whether (X_i, X_j) or (X_j, X_i) is u-q-flippable and calculate $N(F, X_i, X_j, s)$, by checking the clauses in $LCC(X_i, X_j)$ and visiting the entries

$$N(F, X_i, s), N(F, X_j, s), L_1(X_i), L_2(X_i), L_1(X_j), L_2(X_j).$$

We remark that it is unnecessary to check the clauses not in $LCC(X_i, X_j)$, even if the variable X_i or X_j occurs in them, because their contribution to $N(F, X_i, X_j, s)$ has been taken into consideration by $N(F, X_i, s)$ and $N(F, X_j, s)$. Thus, one can update LCQ with the information about whether (X_i, X_j) or (X_j, X_i) is q-flippable and $N(F, X_i, X_j, s)$.

Moreover, because $N(F, X_i, X_j, s)$ is an integer which is generally no more than a few dozen, one can maintain LCQ as several separate sub lists, where the pairs (X_i, X_j) with the same score $N(F, X_i, X_j, s)$ are put in the same sub lists. In addition, each sub list is a queue, where the pairs (X_i, X_j) added first are placed at the front of the queue. Let $LCQM$ be the sub list of LCQ with maximum score. Similarly, we also

maintain the list LCR of critical and reversible pairs (X_i, X_j) and their scores $N(F, X_i, X_j, s)$. Let $LCRM$ be the sub list of LCR with maximum score.

The cost of maintaining the lists of critical and u-q-flippable pairs, as well as critical and reversible pairs, is typically very low, since the expected number of critical variables is constant for random 3-SAT.

4.3.3 Data structure for checking non-critical pairs

For each clause C, we use an element $N(C)$ to record the number of variables that satisfy the clause under the current assignment. It is easy to verify that $N(C_1), N(C_2), \cdots, N(C_m)$ can be maintained within $O(\Delta)$ time for each flip. We maintain the list LN of non-critical clauses C with $N(C) = 1$. For each variable X_i, we maintain the list $LN(X_i)$, which contains all the clauses C in LN where the variable X_i occurs in C.

At any step $t + 1$, assume that the last perform of Line 3 or 4 is at some step $r \leq t$. For each variable X_i, we maintain an element $U(X_i)$. If any variable in $N^+(X_i)$ is flipped in $(r, t]$, then $U(X_i) = 1$, otherwise, $U(X_i) = 0$. We use a list LU to record the list of updated variables X_i with $U(X_i) = 1$.

For each clause C in LN, we also maintain five elements $Q(C), R(C), S(C), P(C), T(C)$. These elements only update when Line 3 or 4 in Algorithm 2 is performed. Recall the definition of u-q-flippable pair in Definition 14. The meaning of these elements is as follows.

- $T(C)$ denotes the step at which $Q(C), R(C), S(C), P(C)$ are updated.
- If any pair of variables occurring in C is u-q-flippable, then $Q(C) = 1$ and $R(C) = 0$. In this case, let $P(C)$ record the u-q-flippable pair with the highest score, and let $S(C)$ record that highest score.
- If no pair of variables occurring in C is u-q-flippable, but there is a reversible pair, then set $Q(C) = 0$ and $R(C) = 1$. In this case, let $P(C)$ record the reversible pair with the highest score, and let $S(C)$ record the highest score.
- If no pair of variables occurring in C is u-q-flippable or reversible, then set $Q(C) = 0, R(C) = 0$. In this case, $P(C)$ and $S(C)$ can be set arbitrary.

Intuitively, $Q(C)$ indicates whether C contains an u-q-flippable pair of variables, while $R(C)$ indicates whether C contains a reversible pair. The pair itself is recorded in $P(C)$, and its score is recorded in $S(C)$. If C contains both an u-q-flippable pair and a reversible pair, only the u-q-flippable pair is recorded. If there are multiple u-q-flippable or reversible pairs in C, only the pair with the highest score is recorded.

For each variable X_i in LU and each clause C in $LN(X_i)$, if the time stamp $T(C)$ is equal to t, we do nothing. Otherwise, we update $Q(C), R(C), S(C), T(C), P(C)$ as follows. Let X_j, X_k, X_ℓ be the variables in C, X_j be the only variable satisfying C and $i \in \{j, k, \ell\}$. Let s_t be the assignment at step t.

- We first update the time stamp $T(C)$ to t.
- Next, we check whether one of (X_j, X_k),(X_k, X_j),(X_j, X_ℓ), (X_ℓ, X_j) is u-q-flippable. Because by Observation 21 we have (X_k, X_ℓ),(X_ℓ, X_k) are not q-flippable. By C is non-critical, we have X_i, X_j do not occur together in any other clause. Thus, we have $N(F, X_j, X_k, s_t) = N(F, X_j, s_t) + N(F, X_k, s_t) + 1$. Similarly, we also have $N(F, X_j, X_\ell, s_t) = N(F, X_j, s_t) + N(F, X_\ell, s_t) + 1$. Assume *w.l.o.g.* that $N(F, X_j, X_k, s_t) \geq N(F, X_j, X_\ell, s_t)$. Consider the following two separate cases:
  - One of (X_j, X_k),(X_k, X_j) is u-q-flippable. We set $Q(C) = 1, R(C) = 0, S(C) = N(F, X_j, X_k, s_t)$. We set $P(C)$ as the u-q-flippable in (X_j, X_k) and (X_k, X_j). The update is done.

- (X_j, X_k),(X_k, X_j) are not q-flippable and one of (X_j, X_ℓ), (X_ℓ, X_j) is u-q-flippable. We set $Q(C) = 1, R(C) = 0, S(C) = N(F, X_j, X_\ell, s_t)$. We set $P(C)$ as the u-q-flippable in (X_j, X_ℓ), (X_ℓ, X_j). The update is done.

- If none of (X_j, X_k),(X_k, X_j),(X_j, X_ℓ), (X_ℓ, X_j) is q-flippable, we check whether one of them is reversible. Because by Observation 21 we have (X_k, X_ℓ),(X_ℓ, X_k) are not reversible. Assume *w.l.o.g.* that $N(F, X_j, X_k, s_t) \geq N(F, X_j, X_\ell, s_t)$. Consider the following three separate cases:

  - One of (X_j, X_k),(X_k, X_j) is reversible. We set $Q(C) = 0, R(C) = 1, S(C) = N(F, X_j, X_k, s_t)$. We set $P(C)$ as the reversible in (X_j, X_k) and (X_k, X_j). The update is done.
  - (X_j, X_k),(X_k, X_j) are not reversible and one of (X_j, X_ℓ), (X_ℓ, X_j) is reversible. We set $Q(C) = 0, R(C) = 1, S(C) = N(F, X_j, X_\ell, s_t)$. We set $P(C)$ as the reversible in (X_j, X_ℓ), (X_ℓ, X_j). The update is done.
  - None of (X_j, X_k),(X_k, X_j),(X_j, X_ℓ), (X_ℓ, X_j) is reversible. We set $Q(C) = 0, R(C) = 0$ and set $P(C)$, $S(C)$ arbitrarily.

The following describes how to determine whether (X_j, X_k) or (X_k, X_j) is u-q-flippable:

- Assume $N(F, X_j, s_t) \geq 0$ and $N(F, X_k, s_t) \geq 0$. By Observation 21, (X_i, X_j) and (X_j, X_i) are both valuable. If (X_j, X_k) is qualified, then (X_j, X_k) is u-q-flippable if X_j is older than X_k, and (X_k, X_j) is u-q-flippable if X_k is older than X_j.

- Assume $N(F, X_j, s_t) = -1$, $N(F, X_k, s_t) \geq 0$, and X_j is older than X_k. By Observation 21, (X_i, X_j) is valuable. If (X_j, X_k) is qualified, then (X_j, X_k) is u-q-flippable.

- Assume $N(F, X_k, s_t) = -1$, $N(F, X_j, s_t) \geq 0$, and X_k is older than X_j. If (X_j, X_k) is qualified, then (X_k, X_j) is u-q-flippable.

In all other cases, neither (X_j, X_k) nor (X_k, X_j) is u-q-flippable.

We maintain LN as separate sub lists. We first use a sub list LNQ to record the clauses C in LC with $Q(C) = 1$ and use a sub list LNR to record the clauses C in LC with $R(C) = 1$ and use a sub list LNO to record the other clauses C in LC where $Q(C) \neq 1$ and $R(C) \neq 1$. Furthermore, we maintain LNQ as several separate sub lists, where the clauses C with the same score $S(C)$ are put in the same sub lists. In addition, each sub list is a queue, where the clauses C added first are placed at the front of the queue. Let $LNQM$ be the sub list of LNQ with maximum score. Similarly, we also maintain LCR as several separate sub lists. Let $LNRM$ be the sub list of LNQ with maximum score.

4.4 The q-flippable or reversible variable with the highest score

When Line 3 or 4 in Algorithm 2 is performed, we first update the data structures as described above. When Line 3 is performed, we pick the first pair (X_i, X_j) in $LCQM$ and the first clause C in $LNQM$ and compare their scores. If (X_i, X_j) has a higher score, then X_i is the q-flippable variable with the highest score. Otherwise, C has a higher score. Then X_k is the q-flippable variable with the highest score where $P(C) = (X_k, X_\ell)$. Similarly, one can also return the reversible variable with the highest score.

4.5 The cost for maintaining the data structures

For the variables, the data structures $N(F, X_1, s), \cdots, N(F, X_n, s)$, $L_1(X_1), \cdots, L_1(X_n)$, $L_2(X_1), \cdots, L_2(X_n)$ can be maintained within $O(\Delta)$ time for each flip. For the clauses, the data structure $N(C_1), \cdots, N(C_m)$ can be also maintained within $O(\Delta)$ time for each flip. For critical pairs, the cost of maintaining the lists of critical and u-q-flippable pairs, as well as critical and reversible pairs, is typically very low.

For non-critical pairs, it is straightforward to show the lists $LU, U(X_1), \ldots, U(X_n), LN(X_1), \ldots, LN(X_n)$ can each be maintained in $O(\Delta)$ time per flip. Updating the list LN is more involved. Specifically, for each flip of a variable X_i, the algorithm determines, for every clause C containing X_i, whether C is in LN. This takes $O(\Delta)$ time overall. Moreover, whenever Line 3 or Line 4 of Algorithm 2 is executed at step $t+1$, each clause $C \in LN(X_i)$ where $X_i \in LU$ requires an update of $Q(C)$, $R(C)$, $S(C)$, $T(C)$, and $P(C)$. Suppose the last time Line 3 or Line 4 was executed was at step $r \leq t$. In that case, at most $t - r + 1$ variables can be flipped in $[r, t]$. Hence, there are at most $(t - r + 1)\Delta$ variables in LU and $(t - r + 1)\Delta^2$ clauses to update. From an amortized perspective, this amounts to a cost of $O(\Delta^2)$ per flip. Nevertheless, even if LN is large and Line 3 or Line 4 is executed frequently, only clauses associated with LU need to be updated, and this list is typically small. Furthermore, once these clauses are updated, retrieving the q-flippable or reversible variable with the highest score takes $O(1)$ time. As a result, the cost of executing Line 3 or Line 4 does not depend on the size of LN, ensuring the overall efficiency of our algorithm.

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